

Branch-wise versus clade-grouped gene-family evolution on the archaeal tree

Reconciliation of 7,059 gene families on the 257-taxon archaeal species tree, Euryarchaeota root

Reconciling each gene family against the species tree under a duplication–transfer–loss (DTL) model requires a choice of how finely rates may vary across the tree. We compared two parameterisations: the *clade-grouped* model, in which all branches of a major clade share a single set of duplication, loss and transfer rates (and origination weight), and a *fully branch-wise* model, in which every branch carries its own rates. As a control, our GPU reconciliation engine reproduces the reference per-family likelihoods to within Pearson $r = 0.999998$, so the comparison reflects the rate model, not the implementation.

Relaxing the clade constraint improves the fit by $\sim 10,900$ log-likelihood units (Table 1) while leaving the inferred root unchanged: Euryarchaeota remains the maximum-likelihood rooting under both models. The improvement is not overfitting — on gene families withheld from rate estimation, the branch-wise model still predicts better than the clade-grouped model (+1,556 units; +0.44 per family), so the additional parameters capture real among-branch rate variation.

That variation has a clear and interpretable structure (Figure 1). Duplication and transfer rates are essentially *clade-constant*: within each named clade the per-branch estimates scarcely depart from a single value (clade membership explains $R^2 \approx 0.94$ of their variance). Loss is markedly less clade-structured ($R^2 \approx 0.38$), with the largest departures along the deep backbone rather than within the well-sampled clades. Biologically, gene gain (duplication, transfer) is well summarised by one rate per clade, whereas gene loss is more branch-idiosyncratic.

The reconstructed ancestral genome is robust to this added flexibility (Figure 2). Of the 7,059 families, the branch-wise model assigns 903 to the last archaeal common ancestor and the clade-grouped model 772; 722 are common to both (Jaccard 0.76), and the per-family probabilities of origination at the root agree closely ($r = 0.96$). The two models therefore recover essentially the same ancestral gene set, the branch-wise model reconstructing a modestly larger genome.

rate model	free params	data log L (nats)	AU 0.05 root set
clade-grouped (one D/L/T per clade)	~ 70	$-1,642,425$	{Euryarchaeota, MHH} (region)
fully branch-wise (per branch)	$\sim 2,000$	$-1,631,537$	{Euryarchaeota}
branch-wise – grouped: +10,888 in-sample; +1,556 on held-out families (cross-validated, not overfitting).			

Table 1: Model comparison at the Euryarchaeota root. Both models recover Euryarchaeota as the maximum-likelihood root; the approximately-unbiased (AU) test supports a {Euryarchaeota, MHH} region under the parsimonious clade-grouped model and narrows to Euryarchaeota alone under the richer branch-wise model — an over-confidence we attribute to model richness, not to stronger evidence.

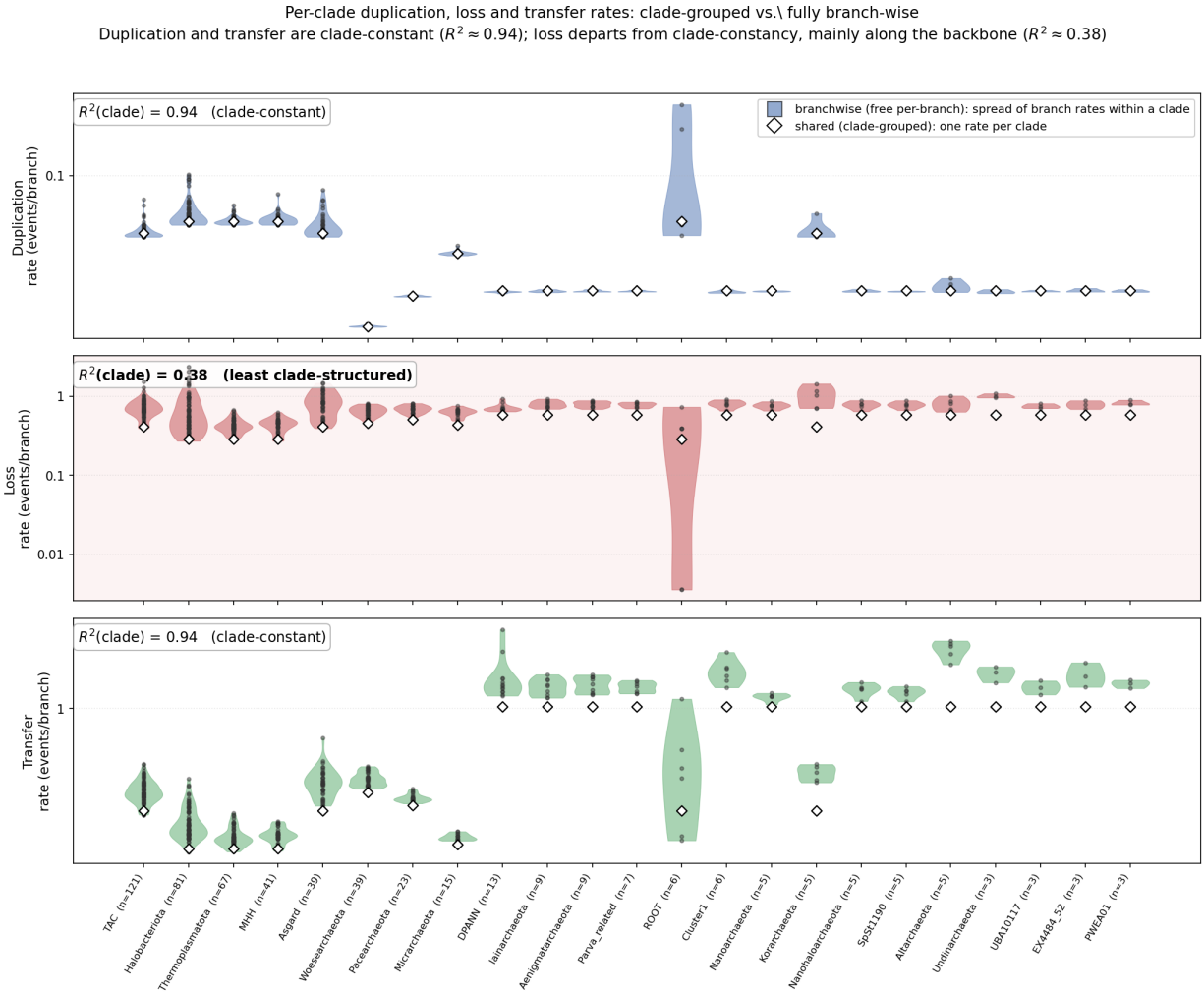


Figure 1: Per-clade duplication, loss and transfer rates (log scale, events per branch): the spread of free per-branch rates within each clade (violins) against the single clade-grouped rate (diamonds). Duplication and transfer are clade-constant ($R^2 \approx 0.94$); loss departs from clade-constancy, chiefly along the backbone ($R^2 \approx 0.38$).

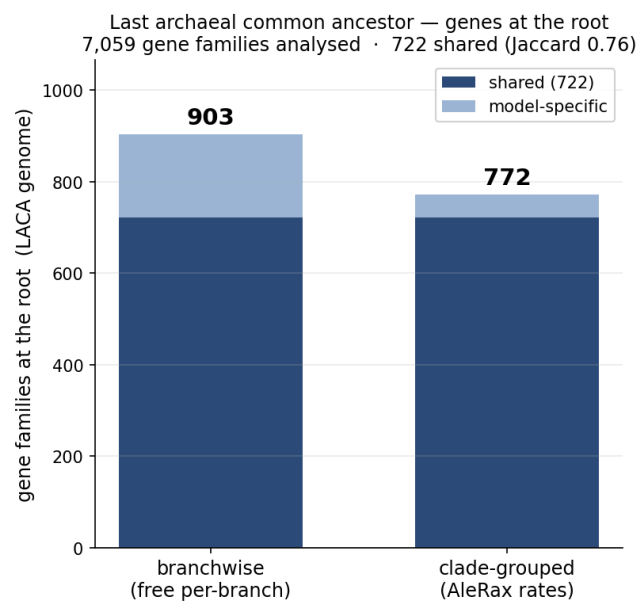


Figure 2: Reconstructed last archaeal common ancestor (LACA) gene content: 903 families under the branch-wise model vs. 772 under clade-grouped rates, 722 shared (Jaccard 0.76), of 7,059 analysed.